King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Stat 201 Major Exam II Term 242

Original exam date: 7 April 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes Original course: STAT 319

---

1. The time to failure (in hours) of a bearing in a mechanical shaft is satisfactorily modeled as a Weibull random variable with $\beta = \frac{1}{2}$ and $\delta = 5000$ hours. What is the mean time (in hours) until failure?

(a) 2215.55

(b) 4431.10

(c) 8862.20

(d) 10000.00

(e) 2500.00

2. Suppose that the random variable X has the continuous distribution given by the following pdf:

$$f(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise} \end{cases}$$

Suppose that a random sample of size $n = 12$ is selected from this distribution. What is the sampling distribution of $\bar{X}$?

(a) The sampling distribution of $\bar{X}$ is exactly $N\left(\mu_{\bar{x}} = \frac{1}{2},\ \sigma_{\bar{x}} = \frac{1}{12}\right)$

(b) The sampling distribution of $\bar{X}$ is approximately $N\left(\mu_{\bar{x}} = \frac{1}{2},\ \sigma_{\bar{x}} = \frac{1}{\sqrt{12}}\right)$, by central limit theorem

(c) The sampling distribution of $\bar{X}$ is approximately $N\left(\mu_{\bar{x}} = \frac{1}{2},\ \sigma_{\bar{x}} = \frac{1}{12}\right)$, by central limit theorem

(d) The sampling distribution of $\bar{X}$ is exactly $N\left(\mu_{\bar{x}} = \frac{1}{2},\ \sigma_{\bar{x}} = \frac{1}{\sqrt{12}}\right)$

(e) The sampling distribution of $\bar{X}$ is approximately $N\left(\mu_{\bar{x}} = 0,\ \sigma_{\bar{x}} = 1\right)$, by central limit theorem

3. The brightness of a television picture tube can be evaluated by measuring the amount of current required to achieve a particular brightness level. A sample of 10 tubes results in $\bar{x} = 317.2$ and $s = 15.7$. Find (in microamps) a 99% confidence interval on mean current required. What is the length of the resulting interval?

(a) 25.5686

(b) 22.4607

(c) 32.2710

(d) 19.4619

(e) 31.4668

4. Suppose that contamination particle size (in micrometers) can be modeled as

$$f(x) = \begin{cases} 2x^{-3}, & x > 1, \\ 0, & \text{otherwise} \end{cases}$$

Determine the mean of X.

(a) 1

(b) 3

(c) $\frac{2}{3}$

(d) 2

(e) $\frac{1}{2}$

5. Of 1500 randomly selected cases of lung cancer, 850 resulted in death within 10 years. Calculate a 95% two-sided confidence interval on the death rate from lung cancer. What is the upper limit of the resulting confidence interval?

(a) 0.5807

(b) 0.6049

(c) 0.5416

(d) 0.5917

(e) 0.6512

6. In a random sample of 85 automobile engine crankshaft bearings, 10 have a surface finish that is rougher than the specifications allow. How large a sample is required if we want to be at least 95% confident that our estimate $\hat{p}$ of the true proportion p was within 0.05 regardless of the value p?

(a) 384

(b) 385

(c) 162

(d) 163

(e) 386

7. In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with a mean of 25 log-ons per hour. What is the standard deviation of the time (in minutes) until the next log-on?

(a) 0.0400

(b) 2.4000

(c) 0.0960

(d) 25.0000

(e) 0.0016

8. The lifetime of a semiconductor laser has a lognormal distribution with $\theta = 10$ hours and $\omega = 1.5$ hours. What lifetime is exceeded by 99% of lasers?

(a) 67846.30

(b) 15.00

(c) 6.51

(d) 725778.39

(e) 668.48

9. In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with a mean of 25 log-ons per hour. What is the interval of time (in minutes) such that the probability that no log-on occurs in this interval is 0.90? (Note: Interval refers to the length of time from start until first log-on)

(a) 0.0921

(b) 2.4000

(c) 0.2529

(d) 5.5262

(e) 0.0042

10. For the copper current measurement, the cumulative distribution function of continuous random variable X is given as:

$$F(x) = \begin{cases} 0, & x < 0, \\ 0.05x, & 0 \leq x < 20, \\ 1, & x \geq 20 \end{cases}$$

What is the probability density function of X?

(a)
$$f(x) = \begin{cases} 0.05x^{-1}, & 0 \leq x \leq 20, \\ 0, & \text{otherwise} \end{cases}$$

(b)
$$f(x) = \begin{cases} 0.05, & 0 \leq x \leq 10, \\ 0, & \text{otherwise} \end{cases}$$

(c)
$$f(x) = \begin{cases} 0.05x^2, & 0 \leq x \leq 20, \\ 0, & \text{otherwise} \end{cases}$$

(d)
$$f(x) = \begin{cases} 0.05x, & 0 \leq x \leq 20, \\ 0, & \text{otherwise} \end{cases}$$

(e)
$$f(x) = \begin{cases} 0.05, & 0 \leq x \leq 20, \\ 0, & \text{otherwise} \end{cases}$$

11. Suppose the current measurements in a strip of wire are assumed to follow a normal distribution with a mean of 10 milliamperes and a variance of 4 (milliamperes)^2. What is the probability that a measurement will exceed 13 milliamperes?

(a) 0.1336

(b) 0.2672

(c) 0.9331

(d) 0.0681

(e) 0.2266

12. In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with a mean of 25 log-ons per hour. What is the probability that there are no log-ons in an interval of 6 minutes?

(a) 0.250

(b) 0.918

(c) 1.000

(d) 0.000

(e) 0.082

13. An electronics company manufactures resistors that have a mean resistance of 100 ohms and a standard deviation of 10 ohms. The distribution of resistance is normal. Find the probability that a random sample of $n = 25$ resistors will have an average resistance less than 95 ohms.

(a) 0.0124

(b) 0.9938

(c) 0.0062

(d) 0.3085

(e) 0.6915

14. A civil engineer is analyzing the compressive strength of concrete. Compressive strength is normally distributed with $\sigma^2 = 1000\,(\text{psi})^2$. A random sample of 12 specimens has a mean compressive strength of $\bar{x} = 3250$ psi. It is desired to estimate the compressive strength with an error that is less than 15 psi at 99% confidence. What sample size is required?

(a) 28

(b) 32

(c) 30

(d) 31

(e) 29

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(d)**	$\mu = \delta\Gamma(1+2) = 5000 \cdot 2 = 10000$.
2	**(c)**	$\mu = \frac{1}{2}$, $\sigma^2 = \frac{1}{12}$; $\sigma_{\bar{x}} = \frac{1}{12}$, approx. normal by CLT.
3	**(c)**	$t_{0.005,9} = 3.250$: length $= 2(3.250)(15.7/\sqrt{10}) = 32.27$.
4	**(d)**	$E(X) = \int_1^\infty 2x^{-2}dx = 2$.
5	**(d)**	$\hat{p} = 0.5667$, SE $= 0.0128$; upper $= 0.5917$.
6	**(b)**	$p = 0.5$: $n = 0.25(1.96/0.05)^2 = 384.16$, so $n = 385$.
7	**(b)**	$\sigma = 1/\lambda = 60/25 = 2.4$ minutes.
8	**(e)**	$z = -2.33$: $x = e^{10-2.33(1.5)} = 668.48$.
9	**(c)**	$e^{-(25/60)x} = 0.9$, so $x = 0.2529$ minutes.
10	**(e)**	$f = F' = 0.05$ on $[0, 20]$, 0 otherwise.
11	**(d)**	$z = (13-10)/2 = 1.5$, upper tail about 0.067; key 0.0681.
12	**(e)**	Mean $= 2.5$ in 6 min: $P(0) = e^{-2.5} = 0.082$.
13	**(c)**	$z = (95-100)/2 = -2.5$, so $P = 0.0062$.
14	**(c)**	$n = (2.576\sqrt{1000}/15)^2 = 29.5$, so $n = 30$.